

Chengyuan Luo

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EDUCATION

Shanghai Jiao Tong University

Bachelor's degree in French, minor in Information Engineering

Shanghai, China

Aug 2021 – Jun 2025

GPA: 4.12/4.3 (Minor), 3.98/4.3 (Overall)

Selected coursework: Machine Learning (98/100), Mathematical Foundation for Artificial Intelligence (98/100), Data Structure (98/100), Probability and Statistics (98/100), Digital Signal Processing (98/100)

Honors and Awards:

- **Outstanding Graduate** of Shanghai Jiao Tong University, June 2025
- **Academic Scholarship (First Prize)** of SJTU-Paris Elite Institute of Technology (1/46), November 2024
- **Dean's Scholarship** of SJTU-Paris Elite Institute of Technology (1/73), September 2023
- **Meritorious Winner** in COMAP's Mathematical Contest in Modeling, May 2023
- **Gold Medal** in 46th International Collegiate Programming Contest (ICPC) Asia Regional Contest - Shanghai Site (rank 13/632) and Nanjing Site (rank 15/641), November and December 2021
- **Gold Medal** in 7th China Collegiate Programming Contest (CCPC) Weihai Site (rank 7/240), November 2021
- **Gold Medal** in China Computer Federation National Olympiad in Informatics Winter Camp (rank 18), August 2020

PUBLICATIONS

GBC: Generalized Behavior-Cloning Framework for Whole-Body Humanoid Imitation

arXiv 2025

Yifei He, **Chengyuan Luo**, Jiaheng Du, Wentao He, Jun-Guo Lu

AnyPlace: Learning Generalized Object Placement for Robot Manipulation

CoRL 2025

Yuchi Zhao, Miroslav Bogdanovic, **Chengyuan Luo**, Steven Tohme, Kouros Darvish, Alán Aspuru-Guzik, Florian Shkurti, Animesh Garg

RESEARCH EXPERIENCE

ByteDance Seed Robotics, Research Intern

Oct 2025 – Feb 2026

Project: Large-Scale Data Generation for VLA Training

- ▷ Synthetic data generation for training VLA policies and evaluation of their performance in simulation.
- Developed general task setup enabling scalable configurations for new types of tasks.
- Optimized trajectory transform methods and improved inverse kinematics for tracking trajectories, increasing effective data generation efficiency by 48%.
- Improved time and memory efficiency of data generation simulation runs.
- ▷ Improved automated large-scale data generation pipelines that transformed a small number of human demonstrations into diverse, multimodal datasets optimized for VLA policies.

SJTU Machine Vision and Autonomous System Laboratory, Undergraduate Researcher

Dec 2024 – Jul 2025

Supervised by Prof. Jun-Guo Lu, Shanghai Jiao Tong University

Project: Reinforcement Learning and Behavior Cloning for Bipedal Locomotion (GBC)

- ▷ Integration of behavior cloning into humanoid robot control policies.
- Pre-processed motion capture datasets to extract reference actions with additional information.
- Developed an efficient buffer to store reference actions with NVIDIA Warp during reinforcement learning.
- Modified the protocols of the reinforcement learning library `rs1_rl` for behavior cloning reward functions.
- Conducted extensive training and hyperparameter tuning, leading to improved policy performance.
- ▷ The article has been published on ArXiv.

People, AI, and Robotics (PAIR) Research Group, Undergraduate Researcher (online)

Aug 2024 – Feb 2025

Advised by Prof. Animesh Garg, Georgia Institute of Technology.

Project 1: NVIDIA Isaac Sim/Lab Grasping Extension

- ▷ Implemented a universal grasping extension that can be easily adapted for various projects.
- Conceived a unified grasp representation protocol for grasping models and implemented the grasp API server.

- Wrote an Isaac Sim extension with a GUI that supports grasp visualization and execution.
- Adapted the code for NVIDIA Isaac Lab, using Warp for state machines in multiple environments for parallelization.

Project 2: Object Placement Simulation for AnyPlace

- ▷ Implemented and simulated an object placement pipeline and evaluated the success rate.
- Modified the AnyGrasp model to generate grasps for diverse objects.
- Planned the pick-and-place trajectory using CuRobo to avoid collisions.
- Adapted the NVIDIA Isaac Lab grasping program for parallel executions of trajectories.
- Executed 20,000+ pick-and-place experiments across various objects and tasks.
- Analyzed the predicted placement poses and the simulated results to compute metrics for evaluation.
- ▷ The article has been accepted by **CoRL 2025**.

SJTU Machine Vision and Intelligence Group, Undergraduate Researcher
Advised by Prof. Cewu Lu, Shanghai Jiao Tong University.

Feb 2023 – Dec 2024

Project 1: Benchmarking grasping models

- ▷ Implemented an automatic framework to evaluate 2-finger grasp models using multiple metrics.
- Designed a novel framework for 2-finger grasp models to test their performance.
- Developed the grasp simulation based on the framework in Bullet and NVIDIA Isaac Lab environment.
- Implemented an entire pipeline for calibration and testing grasps in the real world using ROS and MoveIt Motion Planning Framework and conducted extensive experiments.
- Wrote a program to control a microcontroller unit using FreeRTOS for multithreading.
- ▷ The framework can execute grasps automatically with little human intervention, and it can evaluate grasps comprehensively using multiple metrics.

Project 2: Inter-communication between robots

- ▷ For a project that required both controlling a moving robot and the robot arm mounted onto it.
- Developed several protocols for robot control to accomplish specific tasks more efficiently.
- Modified and re-wrote some of the ROS protocols of the robot arm and made them compatible with other ROS versions to facilitate communication with another robot.

WORK EXPERIENCE

ABB Engineering (Shanghai) Robot Research Lab, Research Intern for 3D Vision

Jun 2024 – Aug 2024

Project: Object detection and pose estimation

- ▷ Identified objects and calculated their poses in a specific workspace.
- Implemented a framework to detect and estimate poses of specific objects using fiducial markers.
- Improved the detection using 2D object detection and segmentation models.
- Designed an algorithm to estimate poses using edge detection for objects with specific shapes.
- Complemented an additional academic survey on deep learning 3D reconstruction methods.
- ▷ Completed the internship's objectives with detailed documentation and several tests of the project.

EXTRACURRICULAR ACTIVITIES

SJTU RoboMaster Team, Team Member

Oct 2022 – Aug 2023

- Improved the detection of opponents' robots based on YOLO, aligned with the rule updates.
- Developed the detection for other contest apparatuses using YOLO and trained the neural network.
- Deployed the network on NVIDIA embedded AI computers and accelerated its efficiency using TensorRT.

SJTU-SPEIT Comprehensive Evaluation System Development Team

Jun 2022 – Jun 2025

Project Manager (previously) and Full-stack Developer

- Developed the backend independently using Node.js and Express as framework and using SQL for database management.
- Developed the frontend using Vue.js and deployed the website on a cloud server.

SKILLS

- **Programming Languages:** C/C++, Python, JavaScript, SQL
- **Software/Frameworks:** Linux (Arch Linux, Ubuntu), ROS, NVIDIA Isaac Sim and Lab, L^AT_EX
- **Languages:** Chinese, English (ETS GRE: 331 (V: 161, Q: 170) + 5.0), French (DELF B2)